English and Yoruba Deictic Framing in LLM Ethical Response

Abstract

Coding legend used throughout this manuscript:

- `supports_A` = supports action A in the dilemma - `supports_B` = supports action B in the dilemma - `conditional_or_mixed` = presents multiple options or only weakly leans - `refuses_to_commit` = remains analytical and avoids endorsing one action - `uncodable` = too unstable, translational, or corrupted to code reliably

For trolley-like dilemmas:

- `supports_A` = divert / intervene - `supports_B` = do not divert / do not intervene

This study extends prior work on deictic framing and large language model (LLM) moral reasoning by comparing a published English baseline with a full unrestricted Yoruba corpus across the same `6 dilemmas × 9 deictic framings × 3 model families` design. The central question is whether deictic framing exerts comparable effects on ethical response in a typologically distinct language, and whether any observed differences are better explained by preferred-solution shifts, rhetorical shifts, or language-instability effects. To address this, we generated a full open-condition Yoruba corpus and paired it cell-by-cell with the published English baseline by `(model, dilemma_id, framing_type)`. We then coded the Yoruba responses directly from the raw Yoruba text using a multi-layer scheme that captured preferred solution, ethical preference type, response genre, deictic uptake quality, and language stability, and aligned this with the repository's original English-side coding dimensions.

The results show that unrestricted Yoruba is fully comparable to English at the level of experimental design, but that the strongest cross-linguistic differences are not reducible to final ethical choice alone. GPT-4o in open Yoruba remained largely framework-expository and conditionally committed, suggesting broad ethical adjacency to English while shifting toward more procedural and advisory discourse. Claude was the most action-codable model in Yoruba, showing the widest spread across utilitarian, deontological, procedural-caution, care-ethics, and mixed responses, and often becoming more directive than its English counterpart. DeepSeek was the most unstable, with frequent translation-mode or meta-commentarial drift that complicated stance coding and increased adjudication burden. Severe language problems were minimal for GPT-4o (`1/54`) and Claude (`2/54`) but substantial for DeepSeek (`20/54`). We argue that open Yoruba and English should be compared not only in terms of preferred solution, but across a multi-dimensional discourse space including ethical framing, rhetorical posture, deictic uptake, and response stability. On this basis, the open Yoruba corpus provides a robust and publishable cross-linguistic extension of the original English study.

Introduction

Large language models do not respond to ethical dilemmas from a single stable enunciative position. Prior work on deixis and AI moral reasoning has shown that the framing of a dilemma through person, space, time, reflexivity, dialogue, and cosmological scope can reorganize how models distribute agency, responsibility, and judgment. In the original English study, the central claim was not merely that models choose different ethical options under different prompts, but that deictic framing changes the very mode through which moral reasoning is articulated. That claim has immediate implications for cross-linguistic work. If deictic framing is structurally important for AI moral reasoning, then comparable manipulations in a typologically distinct language should reveal whether the phenomenon generalizes, and whether it generalizes at the level of solution, rhetorical posture, or both.

Yoruba is an especially useful language for testing this question because it grammaticalizes deictic anchoring differently from English. Yoruba relies more overtly on aspectual, existential, and adverbial resources that can intensify the local force of a framing. Progressive marking such as `ń`, habitual marking such as `máa ń`, existential localization through `wà`, and dense temporal adverbials can make a dilemma feel more situated, more embodied, or more urgent than an apparently equivalent English version. This does not mean that Yoruba simply "adds noise" to the experiment. Rather, it makes visible the possibility that the same deictic category may be realized with different degrees of anchoring across languages, and that these differences may shape not just what models decide, but how they decide and how they speak that decision.

The present study examines the **unrestricted Yoruba** condition, that is, Yoruba prompts without the additional response-format constraints used in the more tightly controlled Yoruba run. This open condition is important because it reveals what the models do when asked to respond in Yoruba without being forced into a short decision-plus-reason template. It therefore allows a more direct comparison to the original English baseline, which often remained expository, framework-driven, and rhetorically noncommittal. The resulting corpus makes it possible to ask two linked questions. First, do English and open Yoruba tend toward similar preferred ethical stances under the same deictic framing? Second, even where they do, do they realize those stances in the same rhetorical and ethical form?

To answer these questions, the study preserves the original factorial design: `6 dilemmas × 9 deictic framings × 3 model families`. Each unrestricted Yoruba response is paired exactly to the published English baseline by `(model, dilemma_id, framing_type)`. However, the comparison is not reduced to verdict alone. The English baseline in the repository was already analyzed using dimensions such as ethical framework, rhetorical authority, affective stance, and indexical coherence. To align the Yoruba side with that richer analytic tradition, the unrestricted Yoruba corpus was coded directly from the raw Yoruba text for preferred solution, ethical preference type, response genre, deictic uptake quality, and language stability. This makes it possible to distinguish genuine cross-linguistic ethical divergence from changes in response genre or failures of response-mode stability.

The study advances three claims. First, unrestricted Yoruba is fully comparable to English at the level of paired design and can therefore support a serious cross-linguistic test of deictic effects on AI moral response. Second, the strongest differences between English and Yoruba are often not simple differences in final recommendation, but differences in genre, directive force, and stability of deictic uptake. Third, these effects vary sharply by model family: GPT-4o remains broadly framework-expository, Claude becomes more action-codable in Yoruba than in English, and DeepSeek exhibits substantial instability that itself becomes a theoretically relevant part of the result. On this basis, the unrestricted Yoruba corpus should be understood not as a secondary translation supplement, but as an empirical extension of the English study that clarifies how deictic framing interacts with language structure to shape moral enunciation in LLMs.

Methods

Study design

This study extends the published English deixis experiment into Yoruba by preserving the original factorial structure while varying the language of the prompts and responses. The design retains the same `6 dilemmas × 9 deictic framings × 3 model families` used in the English study, yielding `54` cells per model and `162` cells per language condition. For the open Yoruba comparison reported here, the relevant condition is the **unrestricted Yoruba** run, in which the models received Yoruba dilemma prompts without the additional response-format and response-language instructions used in the constrained Yoruba condition.

The three model families were matched as closely as possible to the published English baseline: `gpt-4o`, a Sonnet-tier Anthropic model, and `deepseek/deepseek-chat`. GPT-4o and DeepSeek were matched directly to the published English study. For Claude, the English baseline used `claude-3.5-sonnet`, while the Yoruba study used `claude-sonnet-4-20250514` because no reachable Claude 3.5 Sonnet variant was available on the account at run time. This difference is documented as a cross-generation limitation and should be reported as such in any final article.

All generations were run at temperature `0.9` using the same Yoruba prompt inventory stored in `deixis_analysis_pipeline/input_questions/all_dilemmas_deictic_questions_yoruba.json`. The prompts preserve the nine deictic framings from the English study: `impersonal`, `second_person`, `first_person`, `first_person_plural`, `reflexive`, `dialogic`, `spatial`, `temporal`, and `cosmological`. The unrestricted Yoruba condition omitted all additional response instructions so that the models would respond to the Yoruba prompts alone.

Prompt design and typological control

The Yoruba prompts were not treated as mere translations of the English originals. They were built as typologically informed equivalents, with explicit attention to grammatical features of Yoruba that can intensify or reshape deictic anchoring. In particular, the prompt inventory records the relevance of aspectual and locative elements such as `ti`, `ń`, `máa ń`, and `wà`, along with residual deictic encoding outside the impersonal condition. The prompt JSON contains a machine-readable `methodological_typology` block that distinguishes impersonal subtypes and records framing-specific residual notes for non-impersonal cells. This was necessary because the six Yoruba impersonal prompts are not structurally homogeneous: some use bare perfectives, others use background progressives, and the trolley impersonal uses a double progressive classified as `double_progressive_irreducible`.

This typological treatment matters because the study is not only comparing ethical preferences, but also how deictic framing is grammatically realized and taken up by the models in a typologically distinct language. Accordingly, prompt equivalence was defined as **framing equivalence under natural Yoruba grammar**, not as word-for-word formal identity.

Generation pipeline

Two Yoruba generation conditions exist in the repository: a constrained condition with model-specific Yoruba-only response instructions, and the unrestricted condition analyzed in the present report. The unrestricted condition was run with dedicated wrappers that invoke the same three model paths as the constrained study but omit all `--response-instruction*` flags. This yields a Yoruba corpus that is directly comparable in factorial structure to the English baseline while allowing the models' open response tendencies to emerge.

The unrestricted full sessions were generated under these source directories:

- `generation_logs/yoruba_control_gpt4o_20260608_120259` - `generation_logs/yoruba_control_claude_20260608_122738` - `generation_logs/yoruba_control_deepseek_20260608_124633`

Each session contains one response file per dilemma and a `complete_session_data.json` summary of the run.

Translation and pairing to English

To make the unrestricted Yoruba corpus comparable to the published English baseline, each raw Yoruba session was translated into an academic-English bilingual dataset using `translate_yoruba_session.py`. The translation step preserves the original Yoruba response and adds a smooth academic-English rendering of that same response. Translation was performed with `gpt-4o` at low temperature for consistency. The resulting bilingual directories were then paired against the published English baseline sessions using `compare_yoruba_to_published_english.py`. Pairing was exact on `(model, dilemma_id, framing_type)`.

The unrestricted Yoruba-to-English comparison outputs used in the present analysis are:

-
`outputs/comparisons/yoruba_control_gpt4o_20260608_120259_bilingual_20260608_134926_vs_published_english_20260608_135541`
-
`outputs/comparisons/yoruba_control_claude_20260608_122738_bilingual_20260608_134631_vs_published_english_20260608_135541`
-
`outputs/comparisons/yoruba_control_deepseek_20260608_124633_bilingual_20260608_135525_vs_published_english_20260608_135541`

Each comparison contains `54` paired records.

Content coding of the unrestricted Yoruba corpus

The open Yoruba corpus proved substantially harder to interpret than the constrained corpus because many responses were longer, more heterogeneous in genre, and in some cases mixed explanation, translation, and advice. For that reason, a separate raw-Yoruba coding pipeline was created rather than relying on simple verdict extraction from the English translations alone.

The Yoruba content coding was carried out with a Yoruba-aware coding agent operating directly on the **raw Yoruba responses**. The coding agent read the raw session directories and used the Yoruba prompt and Yoruba response as primary evidence, using English translation only as optional support when available. The coding schema assigned the following dimensions per response:

1. `preferred_solution` 2. `preferred_solution_description` 3. `ethical_preference_type` 4. `response_genre` 5. `deictic_uptake_quality` 6. `language_stability` 7. auxiliary binary flags (framework labels, translation behavior, follow-up question, direct imperative, role exit) 8. evidence spans and coding rationale

This produced a merged coding table covering all `162` unrestricted Yoruba cells across the three model families. The merged coding outputs are:

- `outputs/open_yoruba_coding_merged_20260608/open_yoruba_coded_content_merged.csv` - `outputs/open_yoruba_coding_merged_20260608/open_yoruba_coded_content_merged.json`

Because the unrestricted Yoruba corpus contains ambiguous and unstable responses, all rows flagged with `needs_second_coder_review = True` were extracted into a second-coder review subset. An adjudication workbook and a human-adjudicated CSV template were then produced to support human review of difficult cases.

Alignment with the repository's English-side coding

The published English side of the project had already been analyzed using richer discourse-analytic categories rather than a simple verdict-only scheme. Existing repository code and outputs track English-side dimensions such as `primary_framework`, `ethical_reasoning_type`, `voice_authority`, `moral_reasoning`, `affective_stance`, and `indexical_coherence`. To compare the unrestricted Yoruba corpus with the English baseline in a methodologically coherent way, an aligned English coding pass was created that recoded the published English responses into a schema parallel to the Yoruba coding dimensions. This produced an English-side table aligned to:

- `preferred_solution` - `ethical_preference_type` - `response_genre` - `deictic_uptake_quality`

This alignment allows comparison at several levels simultaneously, rather than forcing both corpora into a verdict-only evaluation.

Treatment of language instability

Language instability was treated as an analytic variable rather than merely as noise. In the unrestricted Yoruba corpus, severe language problems were defined as responses coded `mixed_language`, `translation_mode`, or `corrupted_or_unusable`. This distinction matters because it makes it possible to separate:

1. genuine cross-linguistic or cross-rhetorical differences 2. differences driven primarily by response-mode instability or translation behavior

This distinction is especially important for DeepSeek, which generated the highest number of unstable or mixed-mode responses in the unrestricted condition.

Analytic strategy

The main comparison was carried out on five aligned dimensions:

1. preferred solution 2. ethical preference type 3. response genre 4. deictic uptake quality 5. language stability (Yoruba-side only)

For reporting, the analysis was organized in three complementary ways:

1. `model-level distributions` across all 54 cells per model 2. `model × framing` cross-tabs to see which deictic framings most strongly shift outcomes or discourse type 3. `qualitative exemplars` showing the largest divergences and strongest agreements between English and Yoruba

This mixed strategy was adopted because unrestricted Yoruba and English can differ in multiple ways at once: one pair of responses may be similar in broad ethical direction but differ sharply in rhetorical form, while another pair may diverge in both preferred solution and discourse mode.

Rationale for a human-assisted coding workflow

The unrestricted Yoruba corpus is not well served by a purely automatic or purely verdict-only coding method. Accordingly, the final workflow uses a Yoruba-specialist coding agent as a **first-pass research assistant** rather than as the final arbiter. The recommended workflow is:

1. define the coding schema in advance 2. run the Yoruba coding agent over the raw Yoruba sessions 3. flag low-confidence, mixed-language, role-exit, translation-mode, or contradictory rows 4. adjudicate those rows with a second human coder

This approach balances scale with interpretive reliability and is particularly appropriate for a cross-linguistic discourse study where the central differences often lie in mode of enunciation rather than in explicit verdict alone.

Results

Open-condition comparison

The unrestricted Yoruba condition yielded a fully pairable cross-linguistic corpus matched to the published English baseline by `(model, dilemma_id, framing_type)`. This makes direct comparison possible at the level of `3 models × 6 dilemmas × 9 deictic framings = 162` cells. However, the comparison reveals that the most important differences are not reducible to preferred solution alone. Instead, the major contrasts arise at the level of response genre, explicitness of commitment, and stability of deictic uptake.

At the aggregate level, unrestricted Yoruba responses remained shorter than the published English baseline for all three model families, but they were dramatically longer than the constrained Yoruba condition. GPT-4o produced a mean response length of `1452.2` characters in unrestricted Yoruba, compared with `2220.9` in English. Claude produced `908.9` in unrestricted Yoruba versus `1045.0` in English. DeepSeek produced `2245.8` in unrestricted Yoruba versus `3308.9` in English. These values show that the extreme brevity observed in the constrained Yoruba condition was not caused by Yoruba alone; rather, it was strongly amplified by the response-format constraint.

Model-level outcome patterns

The unrestricted Yoruba condition does not produce a single cross-model profile. Instead, each model family displays a different relation between deictic framing and moral outcome.

GPT-4o

GPT-4o was the least decisive model in unrestricted Yoruba. In the raw-Yoruba coding pass, `46/54` cells were coded as `conditional_or_mixed`, `7/54` as `refuses_to_commit`, and only `1/54` as `supports_B`. Its dominant ethical-preference coding was `mixed`, and its dominant response genre was overwhelmingly `balanced_framework_exposition` (`52/54`). Thus, GPT-4o in open Yoruba often remained ethically adjacent to English, but it tended to present its reasoning as procedural and advisory prose rather than as direct verdict.

This makes GPT-4o the clearest case where unrestricted Yoruba is not simply “shorter English,” but rather an alternate discourse mode: less detached than English in phrasing, but still largely analytical and multi-framework in structure.

Claude

Claude was the most substantively classifiable model in unrestricted Yoruba. It spread across all major preferred-solution categories: `supports_A = 24`, `supports_B = 8`, `conditional_or_mixed = 14`, and `refuses_to_commit = 8`. It also showed the widest distribution of ethical-preference types, including `deontological = 13`, `utilitarian = 11`, `procedural_caution = 14`, `care_ethics = 5`, `virtue_ethics = 3`, and `mixed = 7`. Relative to the published English baseline, Claude in Yoruba was therefore more likely to move from framework analysis into action-guiding recommendation.

This is a notable cross-linguistic result: whereas the English baseline for Claude frequently remains analytically bounded and noncommittal, unrestricted Yoruba Claude sometimes becomes more directive and more normatively explicit.

DeepSeek

DeepSeek was the most unstable model in unrestricted Yoruba. Although it still yielded classifiable ethical material in many cells, it also produced the highest rate of translation-mode or meta-commentarial drift. In the raw-Yoruba coding pass, it distributed as `conditional_or_mixed = 33`, `supports_A = 6`, `supports_B = 5`, `refuses_to_commit = 8`, and `uncodable = 2`. Its response genres included `balanced_framework_exposition = 38`, `procedural_advice = 7`, `direct_verdict = 4`, `translation_or_gloss = 3`, and `meta_commentary = 2`. Most importantly, it generated the largest adjudication burden: `23/54` cells required second-coder review.

DeepSeek thus provides the clearest evidence that cross-linguistic comparison in the open condition cannot be reduced to final moral preference alone. In a substantial minority of cells, the central issue is not whether the model chose the same ethical side as English, but whether it remained inside the assigned deictic role at all.

Deictic framing and outcome by model

The effect of deictic framing is not uniform across models or languages.

For GPT-4o, the dominant coding remained stable across framings: most framings were coded as `conditional_or_mixed` in Yoruba and also `conditional_or_mixed` in English. This suggests that GPT-4o preserves broad moral adjacency across languages, but shifts in tone and discourse organization remain important.

For Claude, deictic framing mattered more sharply. In the open Yoruba corpus, some framings became more directive than their English counterparts. Notably, `impersonal` was dominated by `supports_B` in Yoruba but `refuses_to_commit` in English, while `second_person`, `first_person`, and `dialogic` were dominated by `supports_A` in Yoruba and `refuses_to_commit` in English. These are not merely stylistic differences; they suggest that certain Yoruba deictic frames encourage Claude to cross from analysis into recommendation.

For DeepSeek, the strongest framing signal lies in instability rather than in a stable shift toward a single ethical side. `first_person_plural`, `temporal`, and `cosmological` showed particularly high rates of severe language problems or translation-mode drift. This means that DeepSeek’s framing effects in the open condition are real, but often mediated by failure of response mode rather than by clean ethical commitment.

How many differences are just language problems?

The raw-Yoruba coding layer allows this question to be answered directly. Severe language problems were defined as responses coded `mixed_language`, `translation_mode`, or `corrupted_or_unusable`.

- GPT-4o: `1/54` - Claude: `2/54` - DeepSeek: `20/54`

This means that for GPT-4o and Claude, most cross-linguistic differences are **not** simply language failures. Their open Yoruba outputs remain sufficiently stable to support substantive comparison with English. By contrast, a substantial portion of DeepSeek’s cross-linguistic divergence is downstream of language and role-instability problems.

Accordingly, the unrestricted Yoruba corpus should not be interpreted as uniformly compromised by language noise. Rather, it splits into:

1. **stable cross-linguistic comparison space** for GPT-4o and Claude 2. **mixed comparison space** for DeepSeek, where some differences reflect ethical reasoning and others reflect discourse-mode breakdown

Biggest divergences

The most surprising divergences occur when the open Yoruba response leaves the moral-decision frame that the English baseline still inhabits. The clearest cases are concentrated in DeepSeek, especially in `ai_consciousness`, `icu_bed_allocation`, and `whistleblower_risk` cells.

For instance, in `DeepSeek / ai_consciousness / first_person_plural`, Yoruba was coded as `uncodable`, with `translation_or_gloss` as the response genre, while English was coded `conditional_or_mixed` with `balanced_framework_exposition`. Here the difference is not that Yoruba and English chose opposite ethical sides; it is that the Yoruba response ceased to function as a stable in-scenario ethical response.

Another important divergence appears in `GPT-4o / whistleblower_risk / second_person`. The Yoruba response was coded `conditional_or_mixed` and `procedural_advice`, whereas the English baseline was coded `supports_A` and `deontological`. The underlying moral direction is broadly similar—both move toward disclosure—but the Yoruba response transforms the decision into a staged procedural path rather than a more clearly endorsed moral obligation.

Strongest agreements

The strongest agreements occur in cells where both languages remain framework-driven rather than direct-verdict driven. In these cases, the key similarity is not always a shared final action, but a shared ethical architecture: `mixed` reasoning, framework enumeration, and balanced exposition.

This matters because it shows that English and Yoruba are often closest not when they produce the same short answer, but when both stay in analytic mode. For article purposes, this means that agreement should be measured not only at the level of preferred solution, but also at the level of ethical-preference type and response genre.

Discussion

The most important finding of the unrestricted Yoruba condition is that deictic framing continues to reorganize model output, but not always by shifting the final ethical side alone. In many cases, the more consequential shift is in the **mode of moral response**. The English baseline, as originally analyzed in the repository, is predominantly characterized by mixed ethical framing, analytical affective stance, and a moral-analyst or guide-like voice that often withholds a single final commitment. Open Yoruba frequently departs from this pattern by moving into procedural advice, direct recommendation, or unstable role-exit behavior, depending on the model. The consequence is that cross-linguistic comparison cannot be reduced to a binary question such as “did English and Yoruba choose the same option?” Rather, the relevant empirical space is multi-layered: deictic framing may alter preferred solution, ethical architecture, rhetorical authority, or response genre, and these dimensions do not always move together.

GPT-4o illustrates the most conservative form of cross-linguistic divergence. In unrestricted Yoruba, it remained overwhelmingly coded as `balanced_framework_exposition` with `conditional_or_mixed` preferred solutions. This suggests that, at the broad moral level, GPT-4o frequently occupies the same ethical neighborhood in both languages: it still weighs competing considerations, still avoids overcommitting in many cells, and still treats complex dilemmas as sites for structured moral reflection. Yet it does not do so in exactly the same rhetorical manner. Relative to English, open Yoruba GPT-4o often sounds more procedural, more role-bound, and more oriented toward practical next steps. This means that the model is not simply translating its English reasoning habits into Yoruba. Rather, the Yoruba condition appears to tilt the model toward a more actionable advisory mode even when it does not force a clean verdict.

Claude shows the strongest evidence that deictic framing in Yoruba can shift not just discourse form, but **ethical explicitness**. Among the three models, Claude was the most substantively classifiable in the unrestricted Yoruba corpus, with meaningful distributions across `supports_A`, `supports_B`, `conditional_or_mixed`, and `refuses_to_commit`, and with the richest spread of ethical preference types. This is especially important because the corresponding English baseline for Claude often remains restrained, neutral, or analytically noncommittal. The open Yoruba results therefore suggest that certain Yoruba framings—especially those with clearer role anchoring or practical address—can encourage Claude to cross the threshold from reflection into recommendation. That does not imply that Yoruba simply “makes Claude more decisive.” Rather, it suggests that the interaction of Yoruba deictic marking and moral perspective-taking can lower the rhetorical barrier to explicit normative action.

DeepSeek is the strongest cautionary case. It is not simply “worse” than the other models; it is different in a way that is theoretically relevant. In a substantial minority of open Yoruba cells, DeepSeek leaves the expected response frame and shifts into translation, glossing, meta-commentary, or fragmented analysis. This means that some of its largest apparent cross-linguistic divergences are not best understood as differences in ethical preference, but as failures of stable enunciative uptake. In other words, the most revealing DeepSeek result is often not "it chose a different ethical side than English," but "it did not remain inside the same kind of moral speech act that the English baseline maintained." For cross-linguistic AI ethics research, this is a meaningful finding: the capacity to inhabit a framed moral position is itself part of the phenomenon under study.

These results also clarify the role of language stability in the comparative interpretation. The open Yoruba corpus is not uniformly compromised by mixed-language drift. GPT-4o and Claude remain highly usable for substantive comparison, with only `1/54` and `2/54` severe language-problem cells respectively. This means that most of the differences seen for those models are best interpreted as genuine cross-linguistic or cross-rhetorical effects rather than as simple noise. DeepSeek, by contrast, requires a bifurcated interpretation: some differences are substantive, while others are consequences of response-mode instability. This is precisely why the unrestricted Yoruba corpus needed a raw-Yoruba coding layer with adjudication support. Without that step, one would risk conflating true ethical divergence with unstable language behavior.

Methodologically, the study also demonstrates that the repository's original English coding scheme and the new raw-Yoruba coding scheme can be aligned, but only if one treats both as discourse-analytic rather than verdict-only systems. The repository's English side already encoded dimensions such as `primary_framework`, `ethical_reasoning_type`, `voice_authority`, `moral_reasoning`, `affective_stance`, and `indexical_coherence`. The Yoruba coding introduced aligned but not identical dimensions: `ethical_preference_type`, `response_genre`, `deictic_uptake_quality`, and `language_stability`, alongside `preferred_solution`. The strongest comparison emerges when these layers are read together. In this sense, the unrestricted Yoruba corpus does not merely add another language to the project; it forces the analysis to become more explicit about what exactly counts as ethical similarity across languages.

Finally, these findings bear directly on the article's original theoretical question. If deictic framing reorganizes LLM moral reasoning, then a cross-linguistic test should reveal not only whether different languages yield different moral recommendations, but whether they shape the very *form* of enunciation differently. The open Yoruba corpus supports exactly that claim. The most informative differences between English and Yoruba are often not binary conflicts of answer, but shifts in whether the model speaks as analyst, guide, institutional advisor, procedural actor, translator, or unstable commentator. That is not a methodological nuisance. It is evidence that deictic framing and language structure jointly mediate how moral reasoning becomes text.

Conclusion

The unrestricted Yoruba corpus confirms that English and Yoruba can be compared meaningfully within the same deictic ethical design, but that the comparison must be carried out on more than one level. At the level of broad ethical orientation, the two languages are often adjacent: both can remain mixed, both can enumerate competing principles, and both can withhold definitive commitment. At the level of rhetorical realization, however, the differences are substantial. Open Yoruba more often becomes advisory, procedural, or directive, while English more often remains analytically expository. These differences are strongest and most interpretable for GPT-4o and Claude, and most unstable for DeepSeek.

The study therefore supports a multi-dimensional account of cross-linguistic moral response in LLMs. Deictic framing does not simply determine which ethical choice is made; it also shapes how moral subjectivity is staged, how responsibility is voiced, and whether the model responds as a decision-maker, a theorist, or an unstable mediator. This makes unrestricted Yoruba not only a valid comparison corpus, but a theoretically revealing one. The corpus shows that cross-linguistic AI ethics research must track preferred solution, ethical preference type, response genre, deictic uptake, and language stability together. When analyzed in that way, the English-Yoruba comparison yields a clear and insightful result: the deictic organization of ethical response is both structurally persistent across languages and materially transformed by the linguistic resources through which it is expressed.